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First-Aid Diagnosis of a Faulty Personal Computer

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Abstract

First-aid is literally used as emergency medical treatment for persons who are ill or injured and it is given before a more thorough medical attention can be obtained. Whether one is dealing with one's dad's decade-old computer or one's own custom-built gaming rig, troubleshooting computer problems are a part of everyday life. Before making that $50 support call in getting technical support from the expert, one can actually fix it by oneself. This paper uses experts' opinion survey of information technology support engineers on the most common personal computer (PC) problems in the business and how they are fixed. It found that most computer users do not give computer security a second thought. Besides, failure to take some simple steps often results in identity theft or worse problems. It identifies practical directions that can be adopted on a regular basis which can allow computers to run at its optimal speed.

Keywords- *First-aid, diagnosis, faulty, personal computer*

INTRODUCTION

Most computer users often rely on information from the Internet, friends, or computer repair companies for solutions to help their computer perform better. Often, the complaint is either the computer is running so slow or cannot start up normally or its programs could not open or the system has virus. The above descriptions often lead to the purchase of programs that are designed, on the surface, to remedy a wide variety of problems and usually require the services of a computer professional to install and run [1].

The suggested problems could be as a result of:

A. Accumulation of spyware, adware or viruses.
B. The computer user "did something wrong" and/or "deleted something".
C. A download from the Internet could result to some problem.
D. A program which was either incorrectly installed or not completely installed.

While some of these problems may exist on the computer, users can take hope that most of these problems are reversible by using tools that are available within the computer system [1].

Justice (2013) affirms that people are not born knowing how to troubleshoot and maintain computers; they acquire it through experience and exposure to technology. Many professionals in the technical industry learn their trade through long periods of exploration, trial and error, and research in their fields. However, one does not need years of experience to perform basic maintenance chores that can increase the lifespan of one's hardware but only need basic activities that can be done to maintain a computer [2].

The purpose of this paper is to present the basic approaches to solving computer problems by using diagnostic tools that exist on a PC that is using windows operating system.

OVERVIEW OF COMPUTER SYSTEM

Computers work through an interaction of hardware and software. Hardware refers to the parts of a computer that can be seen and touched, including the case and everything inside it. The most important piece of hardware is a tiny rectangular chip inside the computer called the **central processing unit (CPU),** or microprocessor. Sometimes it is referred to as the "computer brain"—the part that translates instructions and performs calculations. Hardware items such as monitor, keyboard, mouse, printer, and other items are often called hardware devices, or devices [3].

Software refers to the instructions, or programs, that tell the hardware what to do. A typical example of it is Microsoft Word used for word processing. The operating system (OS) is also software that manages your computer and the devices connected to it. Two well-known operating systems are Windows and Macintosh operating system because it often comes with the computer system from the factory. 73% of computer users use Windows operating system, 17% use Mac while the rest use Linux operating system. It is the operating system that control other programs installed on a system. The inter relationship of hardware, operating system, application software, and user is shown **in Figure 1**

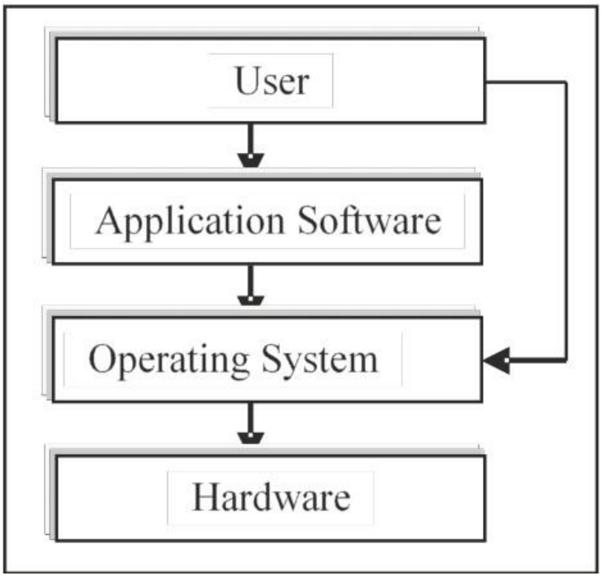

Figure.1: Operating system as intermediary between Application software and hardware.

The three major works of operating system in computer are as follows:

A. *Allocation of resources:* The OS determines what computer resources (CPU, Radom Access Memory (RAM or memory), disks, input and output devices) a program would need and allocate it to run the program.

B. *Schedule of operations:* The OS is responsible for scheduling programs so as to use system resources efficiently.

C. *Monitoring system activities:* Some of the important activities of the Operating System (OS) include:

1) Security: By means of password and similar other techniques, preventing unauthorized access to the programs and data.
2) Control over system performance: Recording delays between request for a service and response from the system.
3) Job accounting: Keeping track of time and resources used by various jobs and users.
4) Error detecting aids: Production of dumps, traces, error messages and other debugging and error detecting aids.
5) Coordination between other software and users: Coordination and assignment of compilers, interpreters, assemblers and other software to the various users of the computer systems [4, 5].

COMMON FAULTS IN PERSONAL COMPUTER

There are thousands of problems that PC could have, from an endless list of possible error messages to various hardware failures. Research shows that even the most stable, secure machine is not safe from buggy software, failing hardware, or even the occasional loose wire [6, 7]. Most of those problems may fall within these categories:

A. *Lockups and Freezes:* This occurs when computer halt and displays an error message that is utterly indecipherable to most computer users. Lockups can be caused by buggy software, or a system running out of memory.

B. *Slow Performance:* This could occur as a result of accumulated useless files or applications deleted from the computer. These files form all kinds of junk thereby degrading system performance. Ironically, antivirus and anti-malware software, while vital to the health of a computer, can slow down a PC's performance. Low hard drive space or not enough RAM in the machine can also cause computer errors and slow down the machine.

C. *Strange Noises:* A lot of noise coming from one's computer as a result of either hardware malfunction or a noisy fan constitutes a strange noise.

D. *Overheating:* This occurs when computer's components generate excessive heat during operation.

E. *Slow Internet:* Slow Internet browser performance occurs usually when browsers cookies is filled up.

F. *Dropped Internet Connections:* Dropped Internet connections can be very frustrating. It could be as a result of viruses, a bad network card or modem, or a problem with the driver.

G. *Windows not Booting:* This occurs when computer suddenly shuts off or has difficulty in starting. It could have a failing power supply.

H. *Abnormally Functioning Operating System or Software:* This occurs when the operating system or other software is either unresponsive or is acting up.

These and many more are the various faults or problems that are usually faced by computer users. The next section of this paper explains the basic first-aids as a means of preventive maintenance inspection to preserve the reliability of the personal computers (Pcs).

BASIC FIRST-AID MAINTENANCE FOR FAULTY PERSONAL COMPUTER

A. *Perform Windows Update:* It is advisable for one to stop the usage of computers with insecure or outdated operating systems. Anytime one uses computer with old versions of Microsoft windows, one is putting it at risk. These updates repair certain bugs discovered after time, reduce the security vulnerability of your system and add functionality to one's operating system [8, 9].

B. Use anti-spyware and anti-virus software: An accumulation of spyware can severely slow down one's PC, leading it to crash, or behave abnormally. Antivirus software works based on known viruses. Most anti-virus programs run in the background and scan all incoming e-mails, downloaded files, etc. Hence, it is not required to run a scan unless you suspect your computer is infected. To be on the safe side, run a full system scan once a month [8, 9].

C. Perform Disk check: This procedure determines the integrity of hard drive. It verifies files & folders on the hard drive. While it is checking, it will describe any actions it takes and the windows closes once it is finished. Disk check should be done once in every three months [1, 9].

D. Removal of unnecessary Programs: Many computers store a large number of programs which are never used. These programs can be removed from control panel. However, any programs that start out with “Windows” should be left alone and also any programs that refer to equipment you have installed or your Anti-virus program should be left alone. If you are not sure as to which are the equipment programs, do not remove it [1, 8].

E. Disk Defragmentation: This procedure rearranges the files on the hard drive so that they are accessed more efficiently by the computer. Over the long run, a disorganized hard drive can eventually crash and obliterate all stored data [1, 9].

F. Use a firewall: A firewall simply tries to block hackers from entering or using one's computer. As a frequent Internet user, it is imperative that one have the Windows Firewall enabled to protect one's computer from hackers [8].

G. Hardware Upgrades: When computers are running slow, much of the problem can be traced to a Random Access Memory (RAM) overload. Memory controls the speed and the accessibility of user actions and programs running (simultaneously) on the computer. Whenever a program or a piece of equipment is installed, the software that is installed with it places a terminate and stay resident (T.S.R.) program that can be memory resident. This means that it stays and stored in memory. Therefore, the more programs and equipment you have, the more memory is being used. The ways by which one can prevent this is by upgrading the memory equipment to higher size or 'free-up' the memory. [1, 9]

H. Power laptops down before travelling: Hard drive failure can occur when moving parts get jammed and out of alignment. Make sure to power down laptops before you slip them in a bag and head to a new location. This prevents jostling of moving parts while the computer is on. Computers can also overheat in a backpack or purse, if they are not properly switched off or put to sleep. Disasters can be avoided if one always power down one's computer before commuting especially laptops [2].

I. Do not keep battery-powered devices plugged in: Many people make the mistake of leaving their electronic devices constantly plugged in, even when the battery is full. This is the fastest way to kill it. The best way to maintain a battery is to charge it up, unplug it, and use the power until it is empty [2].

J. Create backups: Diskettes are no longer in use for backup - a Compact Disk Recordable (CD-R) drive can help quickly backup important data (700 MB per disc or equivalent to 485 diskettes). Digital Video Disk (DVD) recordable drives are also available (~7 times as much as a

CD or equivalent to 3200 diskettes) Other options include external Universal Serial Bus (USB) hard drives to store all data- documents, photos, music. Cloud service can also be used in backups [2, 8].

K. *Use complex passwords:* User error can quickly lead to computer problems, especially if several people share a machine. Whether at work or at home use complex passwords (and never write them down). Using a password longer than 8 characters can greatly reduce the chance that someone will guess your password [2, 8].

L. *Enlist the support of experts:* All these maintenance procedures can be scary. Hackers and even unexpected problems with security patches could potentially mess up one's system rendering it unusable. It may also be time consuming to update or download and installations can take hours. It is not a crime to enlist the assistance of experts. There are many companies that specialize in providing home user support [8].

CONCLUSION

This paper presented basic steps for troubleshooting and solving PC problems and faults. It features the common windows-based computer faults and the preventive maintenance inspection from support engineers to reduce maintenance costs of PC and increases its overall performance.

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